

Threadline

Your app, on *iMessage*.

Paste a store URL. Get an AI agent that knows the business, is tested on simulated customers, and answers your customers in blue bubbles.

Live: threadline.akshitbansal1313.workers.dev

Code: github.com/Reppin123/threadline

[AKI: your name + one line about you, in your own words]

do you have something for diwali gifting?

The Kangra Trio Gift Box ₹999, or The Chai Ritual Set ₹549. Want me to place an order here?

PROBLEM

Customers answer texts. Brands talk to them through *widgets and email.*

THE CHANNEL GAP

5.6% **1.69%**

SMS campaign click rate email campaign click rate

Text gets about 3x the clicks of email. Yet support and sales still live in a chat bubble on a website the customer has already left.

WHY BRANDS DON'T FIX IT

A text agent that is safe to put in front of customers needs five hard parts:

- read the whole site and catalog, keep it fresh
- take real actions (orders, bookings) through tools
- remember each customer across conversations
- test it before a customer sees a wrong price
- iMessage plumbing, delivery, retries, routing

WHY NOW

iMessage just became *programmable*, and the US is an iPhone market.

1 · THE PIPE OPENED

\$0 to start

Photon Spectrum is open source (MIT) and its cloud gives managed iMessage lines: free up to 10 users, \$25/mo up to 100, \$250/line/mo for a dedicated number.

2 · THE AUDIENCE IS THERE

69%

iPhone share of US smartphone sales in Q4 2025, a record high (65% a year earlier). Apple added RCS to Messages in iOS 18, so non-iPhone customers get rich fallback too.

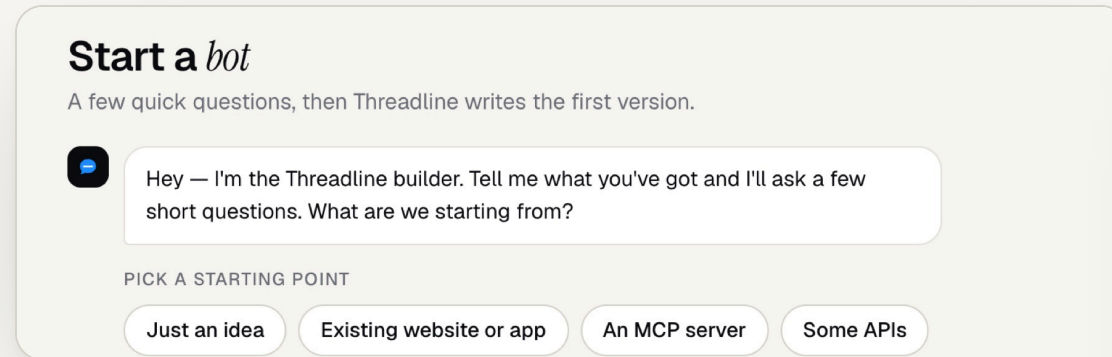
3 · AGENTS CAN ACT

3.8s_{p50}

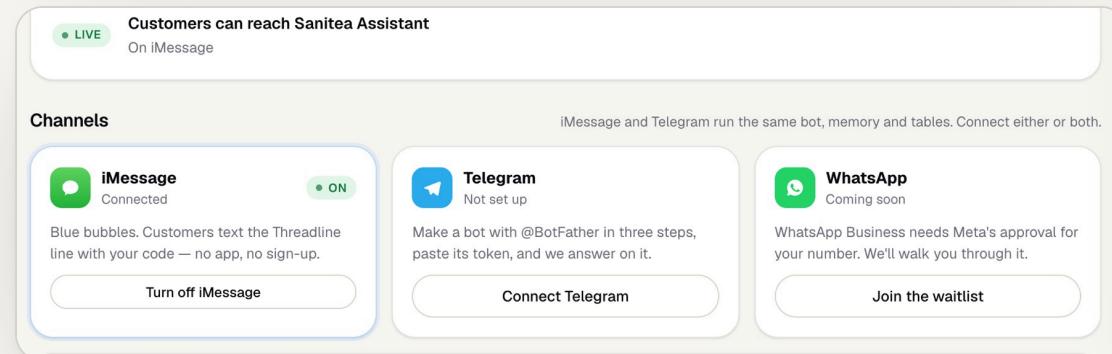
LLM tool use is now reliable enough to take an order in chat. Our Sanitea bot read 32 pages, saved an order to a table and answered at 3.8 s median on claude-sonnet-5-5.

PRODUCT

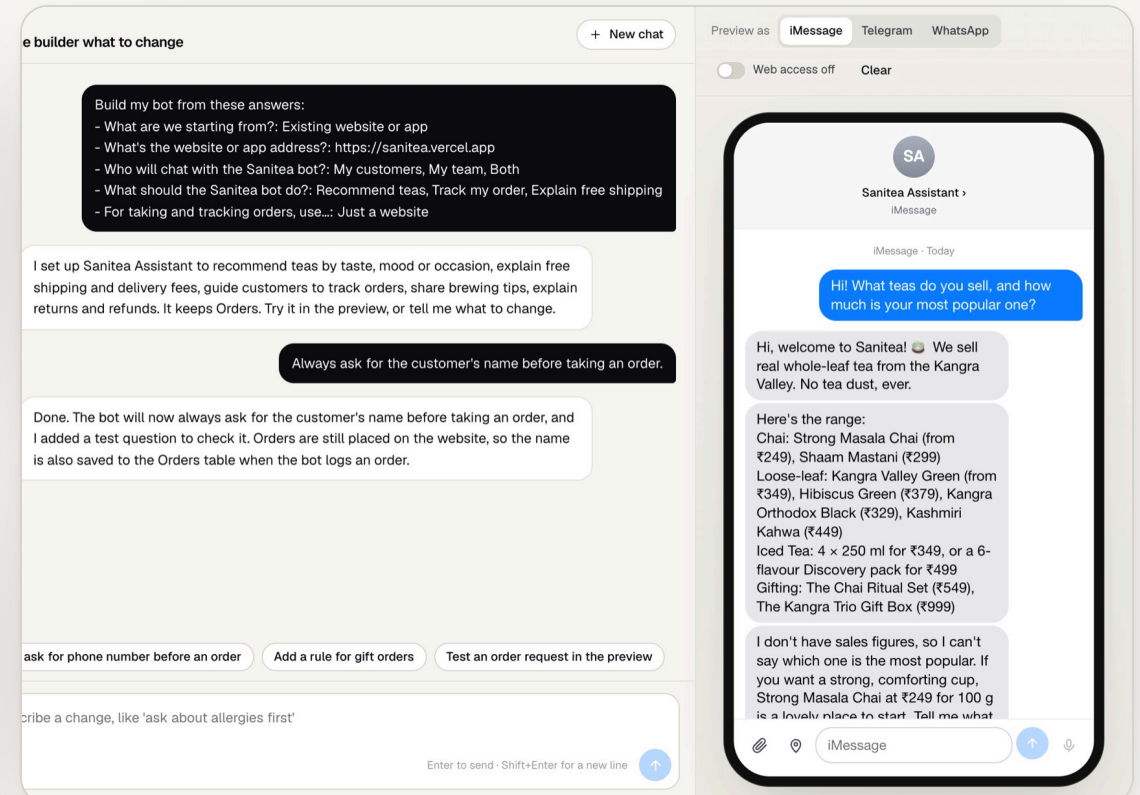
A URL in, a tested iMessage agent out. *About two minutes.*



1 • Start. A website, an OpenAPI or MCP server, or just an idea.



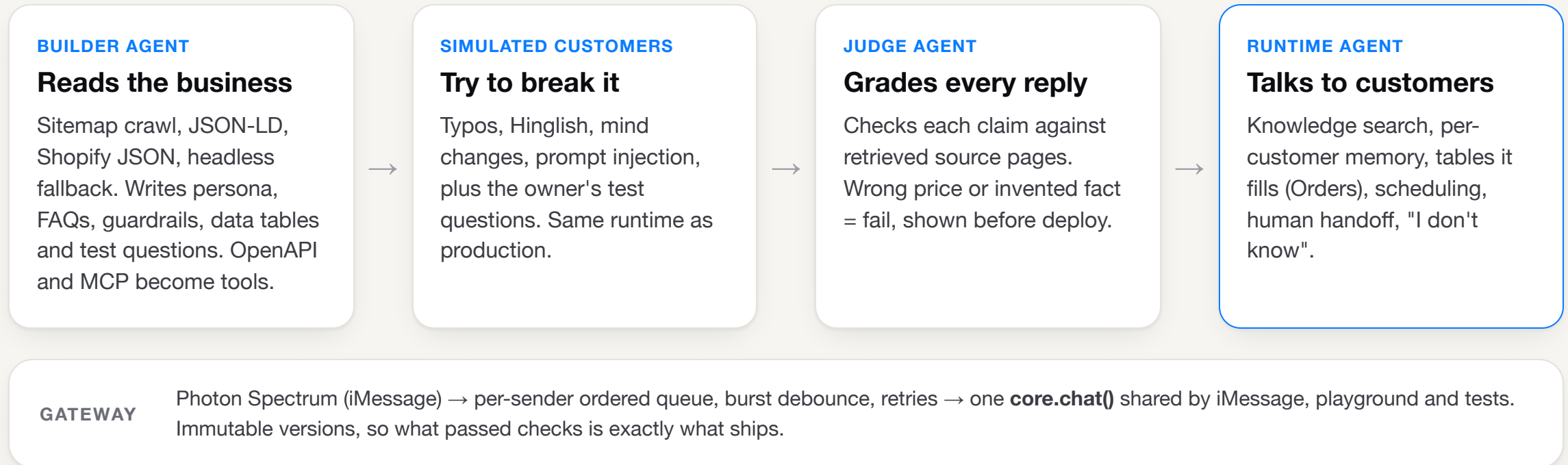
3 • Go live. Deploy a version to iMessage, Telegram next. Roll back any time.



2 • Shape it. Chat with the builder; the iMessage preview answers with real prices from the site.

HOW IT WORKS

Four agents behind every bot, so *the customer never meets a first draft.*



PROOF, MEASURED IN THIS BUILD

It works on a real store, *and we can show the receipts.*

12/12

end-to-end checks on
sanitea.vercel.app: prices,
gifting, brewing, shipping, order
saved, memory across chats,
Hinglish, "don't know"

96%

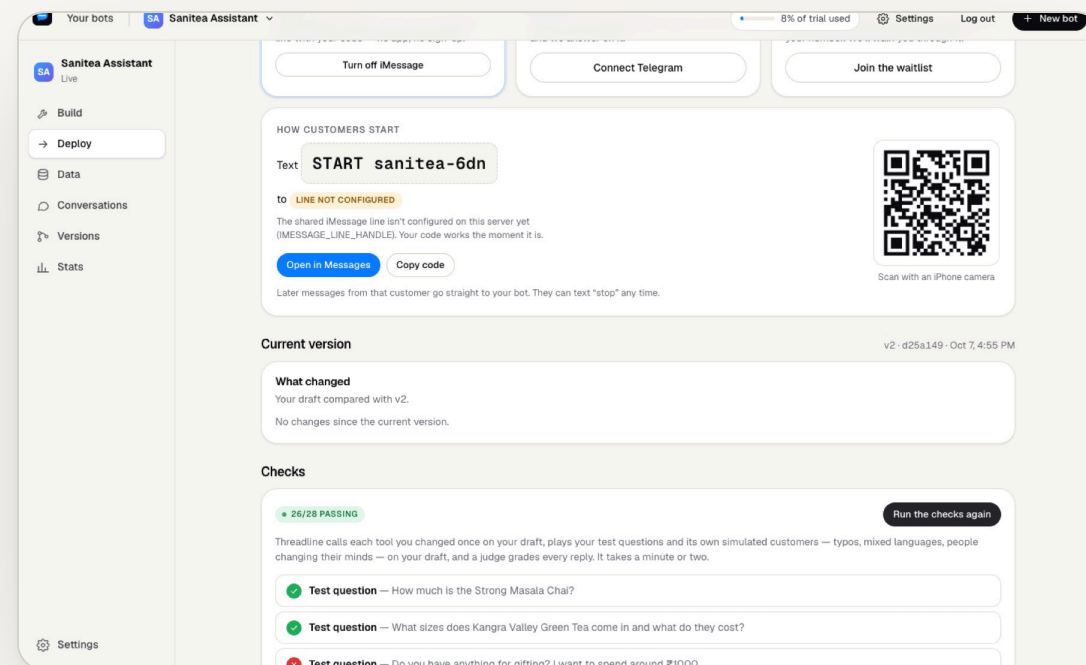
simulated-customer checks
passing (25/26). The deployed
bot shows 26/28.

3.8s

median reply (3.1 to 5.7 s across
runs) on claude-sonnet-5-5

19

tools generated from the
Petstore OpenAPI; a live
getPetById call succeeded. An
idea-only bakery bot passed 2/2
on mock data.



Deploy page: checks run on every version before it goes live.

HONEST STATUS

0 paying customers yet. Gateway is connected to Photon Spectrum Cloud; the first real-phone send was blocked by the free plan's allowlist, fix is registering the handle in Photon.

An orchestrator and seven specialist agents shipped it in *84 minutes*.

AGENT	OWNS	SHIPPED	COMMITTS
orchestrator	spec, contract, schema	frozen contract.ts, job queue, priorities	6
core	packages/core	ingest, runtime, memory, checks, judge	16
gateway	apps/gateway	Photon iMessage, routing, outbound	14
web	apps/web	site, wizard, builder, deploy, API	11
platform	worker, deploy	job worker, supervisor, smoke test	8
supabase	persistence	SQLite snapshots to Supabase Storage	2
telegram	new channel	Bot API poller, 35/35 simulator tests	1
pitch	pitch/	this deck, screenshots, PDF	1

HOW THEY WORK TOGETHER

One frozen interface (contract.ts), strict file ownership, and an append-only coordination log with 20 dated hand-offs. No human wrote code.

A REAL HAND-OFF, 16:25 TO 16:30

Gateway found Photon's free plan has no shared inbound number. Orchestrator switched onboarding to "bot texts you first". Gateway shipped the outbound path 5 minutes later, with tests.

First commit 15:43 → live on Cloudflare 17:07. ~13,000 lines of TypeScript.

Start with Shopify brands that *already pay* for support or SMS software.

WHY THIS ICP FIRST

- A store URL is all we need: catalog, prices and policies are machine-readable (our Shopify JSON path).
- They already have a budget line for answering customers: helpdesks and SMS tools.
- Questions are repetitive and checkable: price, shipping, returns, order status.
- We can show a working bot before the first call.

BEACHHEAD · SHOPIFY STORES ON GORGIAS OR POSTSCRIPT

$$109,648 \times \$2,400/\text{yr} = \$263\text{M}$$

69,694 Gorgias + 39,954 Postscript installs. Global and may overlap, so an upper bound.

EXPANSION · EVERY LIVE US SHOPIFY STORE

$$1,186,833 \times \$1,188/\text{yr} = \$1.41\text{B}$$

At the \$99/mo minimum plan.

\$2,400/yr = 400 conversations a month × \$0.50, our proposed price (next slide).

BUSINESS MODEL

Metered credits: *\$0.50 a conversation*, half what helpdesk AI charges.

PER CONVERSATION (5 REPLIES)		\$
LLM cost per reply, claude-sonnet-5-5	measured	0.011
× 5 replies per conversation	assumption	0.055
iMessage line: \$250/line/mo <i>sourced</i> , shared by 20 brands × 1,000 conv.	assumption	0.013
COGS		0.068
Price	proposed	0.50
Gross margin		86%

PRICING

Free \$1.50 credit to build and test (live today). Then \$0.50 per customer conversation, \$99/mo minimum. Own dedicated number: +\$350/mo on Photon's \$250 line.

WHAT IT REPLACES

Gorgias AI Agent: \$0.90 to \$1.00 per automated interaction. Intercom Fin: \$0.99 per outcome. Both answer in a widget or inbox, not in the customer's texts.

ONE-TIME COST TO ONBOARD A STORE

Build ~\$0.11 + full checks run ~\$1.52 = ~\$1.63 of LLM spend, measured.

We build their bot *before* we ever email them.

List Gorgias / Postscript stores, StoreLeads	1,000 stores / month	~\$300 data
Pre-build + test bot from their URL, checks run	1,000 tested bots	\$1.63 each = \$1,630
Founder email "Your store's bot is live, text it"	150	15% try it
Trial connect their number, real customers	30	20% start trial
Paid \$200/mo average	12	40% convert

UNIT ECONOMICS

CAC $\approx$ $(\$300 + \$1,630) / 12 \approx$ **\$160**

Gross profit $\approx$ $\$200 \times 86\% =$ \$172/mo

Payback $\approx$ **1 month**

PROOF THE MOTION WORKS

Step 2 is already automated: the Sanitea bot in this deck was built from a URL with no human input, in 99 s.

WHAT WE VALIDATE IN PHASE 2

The three conversion rates above are targets, not results. First 1,000-store batch tells us the real ones.

Three things that are hard to copy *together*.

MESSAGE FIRST

Built for the blue bubble

Shared-line routing by join code, "bot texts you first" for Photon's pooled numbers, ordered per-sender queues, burst debounce. Telegram on the same brain, 35/35 simulator tests.

TESTED BEFORE LIVE

No first drafts in front of customers

Every version runs simulated customers and a judge against the source pages. Immutable versions mean what passed is what ships, with one-click rollback.

DONE FOR YOU

A URL, not a project

No decision trees or intents to configure. The builder reads the store and writes the bot; changes are a chat message. That is also what makes our pre-built outbound funnel possible.

And a cost edge: the company itself is run by agents, so a new channel or vertical is a brief and an afternoon, not a hiring plan. Telegram went from brief to 35 passing tests the same day.

From a working product to *first paying brands*.

USERS

Run the funnel on a first batch of Shopify stores; get real customers texting real brands.

CHANNELS

Real-phone iMessage via Photon, Telegram live on Deploy, WhatsApp waitlist.

PLATFORM

SQLite snapshots today, Postgres on Supabase next; credits billing.

LINES

A dedicated Photon Business line so customers can text a number first.

LIVE TODAY

12/12 E2E ON A REAL STORE

96% SIMULATED CHECKS

3.8 S MEDIAN REPLY

84 MIN, FIRST COMMIT TO LIVE

The ask: [AKI: amount and what it buys, in your own words]

Try it: threadline.akshitbansal1313.workers.dev

Code: github.com/Reppin123/threadline

One container, three processes, *one brain*.

APPS/WEB · NEXT.JS 15

Landing and SEO pages, auth (magic link, password, Google), wizard, builder, deploy, data, conversations, stats, public API with keys.

APPS/GATEWAY · PHOTON SPECTRUM

iMessage cloud / local Mac / terminal modes, Telegram long polling, line_routes binding, per-sender queue, outbound scheduler, retries.

APPS/WORKER

Durable job queue: build_bot, run_checks, recrawl. Concurrency 2, stale-job requeue, heartbeat.

PACKAGES/CORE

LLM provider abstraction, ingest (site, OpenAPI, MCP, idea), FTS5 retrieval, tools, memory, runtime loop, checks, judge, insights.

PACKAGES/DB

SQLite (node:sqlite, WAL), migrations, job queue, AES-256-GCM credential encryption, snapshots to Supabase Storage every 30 s.

HOSTING

Live on Cloudflare Containers. Same Docker image runs on Fly or Render. Scale path: Postgres + pgvector behind the same db helper.

Everything a bot knows, does and says is *a row you can audit.*

AREA	TABLES	WHY IT MATTERS
Accounts	users, sessions, api_keys (sha256), usage	Metered \$ per category is the billing base
Bots	bots, versions (immutable snapshots), channels	What passed checks is exactly what ships; rollback
Knowledge	knowledge_docs, chunks + FTS5, tools	Grounded answers, cited by the judge
Customers	customers, memories, conversations, messages	Per-customer memory across channels and threads
Actions	bot_tables, rows (e.g. Orders), scheduled_messages	The bot does work, not just talk
Quality	test_runs, test_cases, events	Pass rate per version, insights
Routing + jobs	line_routes, jobs	Shared iMessage line, durable background work

Briefs, ownership, a contract and a log.

SHARED RULES (AGENTS/COMMON.MD)

- Hard folder ownership; contract edits additive only
- Schema changes as new numbered migrations only
- Cross-agent needs go in COORDINATION.md with a stopgap
- Loop: build, typecheck, run for real, fix; status file after each milestone
- Never print or commit secrets; keys loaded from Keychain

EACH BRIEF DEFINES

- Scope and files owned
- A Definition of Done, ticked only when proven by a run
- Exact test commands (e2e, simulator, smoke)
- "Needs Aki" list for things only a human can do (accounts, phone)

Verification is real: web e2e 16/16 in headless Chrome, gateway simulator 24/24, worker smoke PASS, core e2e 12/12.

Every number, and where it came from.

FIGURE	SOURCE
1,186,833 live US Shopify stores (38.2% of 3,103,755)	StoreLeads, storeleads.app/reports/shopify, updated Oct 2 2026
69,694 Gorgias · 39,954 Postscript Shopify installs	StoreCensus, storecensus.com/stats/app/gorgias and /postscript-sms-marketing, Oct 2026
Gorgias AI Agent \$0.90 (annual) / \$1.00 (monthly) per automated interaction	gorgias.com/pricing, checked Oct 2026
Intercom Fin \$0.99 per outcome	fin.ai/pricing, checked Oct 2026
Photon: free 10 users, Pro \$25/mo 100 users, Business \$250/line/mo	photon.codes/pricing (research/photon/pricing.txt)
iPhone 69% of US smartphone sales, Q4 2025	Counterpoint Research (Feb 2 2026), reported by TechSpot
RCS in Messages, iOS 18	Apple Newsroom, Jun 10 2024
SMS 5.6% vs email 1.69% campaign click rate	Klaviyo 2026 SMS and email benchmarks
12/12 e2e, 96% checks, 3.8 s p50, 19 OpenAPI tools, \$/reply, \$/build	This build: AGENTS/core-e2e-transcript.md, STATUS-core.md, usage + test_runs tables